

Advanced malware and their impact on virtualization: A case study on hybrid feature extraction using deep memory introspection

Mohit Bhatt
Doon University, Dehradun
Dehradun, Uttarakhand, India
mohitbhatt713@gmail.com

Avantika Gaur
Doon University, Dehradun
Dehradun, Uttarakhand, India
avigaur99@gmail.com

Saksham Badoni
Doon University, Dehradun
Dehradun, Uttarakhand, India
sakshambadoni472@gmail.com

Yutika Agarwal
The NorthCap University
Gurgaon, Haryana, India
agarwal.yutika@gmail.com

Preeti Mishra*
Doon University, Dehradun
Dehradun, Uttarakhand, India
dr.preetimishranit@gmail.com

ABSTRACT

As pandemic has hit the world, virtualization has become the hot topic of today's era. Almost every organization has shifted to the virtual environment (specially cloud computing). However, the security concerns in virtualization are the central issue for the researchers as well as organizations. Attackers use different tactics to exploit the vulnerabilities present in virtual components. In this paper, we provided a detailed study on the malware families & along with their impact on virtualization. In addition, malware log extraction using deep memory introspection has been explored. Various plugins have been explained, along with the variety of features that are essential for malware analysis purposes. A case study has also been provided using the testbed set up in our lab to provide the practical insight on deep memory introspection using open source tools such as LibVMI, DRAKVUF, etc., along with their usage to extract different features outside the VM at the hypervisor. We hope that our work will help readers to understand the malware logs and extract important features for malware analysis in a virtual environment.

ACM Reference Format:

Mohit Bhatt, Avantika Gaur, Saksham Badoni, Yutika Agarwal, and Preeti Mishra. 2022. Advanced malware and their impact on virtualization: A case study on hybrid feature extraction using deep memory introspection. In *2022 Fourteenth International Conference on Contemporary Computing (IC3) (IC3-2022)*, August 4–6, 2022, Noida, India. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3549206.3549223>

1 INTRODUCTION

With the evolution of new threats and vulnerabilities, malware analysis has become a crucial challenge in virtualization security.

*Corresponding author

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

IC3-2022, August 4–6, 2022, Noida, India

© 2022 Association for Computing Machinery.

ACM ISBN 978-1-4503-9675-2/22/08...\$15.00

<https://doi.org/10.1145/3549206.3549223>

As per the National Institute of Standards and Technology (NIST), the National Vulnerabilities Database (NVD) has recorded 2019 vulnerabilities in February 2022¹. Moreover, with the addition of a virtualization layer in the modern era, the attacking vulnerabilities have increased. Attackers target different virtualization components by exploiting the vulnerabilities. IBM Security X-Force Threat Intelligence [11] released a report in which various cloud attacks have been discussed.

Virtualization consists of Virtual machines (VMs), virtual monitor machine (VMM/ hypervisor), virtual networks, the host OS, and underlying hardware. A virtual machine (VM) is an application (that acts like a computer) that offers a virtual environment in which to execute an operating system (OS). A hypervisor is a thin layer at the top of the host OS or over the hardware that offers the abstraction of the hardware. It runs multiple VMs in the underlying hardware. A VM can be a potential target of an attacker [9]. Attackers can use worms like Vobfus, password stealers like Zbot, or backdoors trojan like Cycbot.G, Robot to steal sensitive data and tamper with the functionality of guest OS. If VM isolation is not implemented correctly, a co-hosted infected VM can carry out different attacks on other VMs [7]. The security concern regarding virtualization has been a hot topic of researchers for the last few years [8].

Malware analysis is one of the most important aspects of virtualization security. There are different approaches to analyzing malware. The static malware analysis examines a malware file without actually running its code. The information that can be obtained from the static analysis includes source code, byte code, file attributes, opcodes, etc. The dynamic analysis examines the run time characteristics of malware in the controlled environment [6]. The dynamic analysis features include system calls, instruction traces, memory traces, and so on. A hybrid analysis is one which merges static and dynamic analysis [14]. It retrieves the static features of malware first, then combines them with the malware's dynamic features for a comprehensive analysis. Malware detection is a subcategory of malware analysis in which various methods are used to identify a threat, prevent infections and find the solution for the same [1]. There are two types of malware detecting software. The first is a signature-based strategy, in which threats are identified based on a pattern. The second type is behavior-based or

¹<https://nvd.nist.gov/vuln/full-listing/2022/2>

anomaly-based detection, which compares malware activity against a collection of clean templates [15].

In this paper, a comprehensive study is provided on various malware families, their characteristics, and their impact on virtualization. For instance, *Lolyda* family is a password stealer malware that steals users' credential and sends it to the remoter server. Furthermore, the paper describes various advanced dynamic analysis features extracted via deep memory introspection of programs running inside the VM. More specifically, the use of various open-source introspection tools and libraries such as *DRAKVUF*² along with plugins (such as SYSCALL, SYSRET, PROCMON, FILETRACER etc.), *Volatility*³, *LibVMI*⁴ etc. is discussed to extract distinct dynamic features. Next, a case study is provided to demonstrate the steps of hybrid feature extraction using open source tools. The malware dataset is collected from San Jose State University for analysis and is injected from the host to VM. The significant contributions of the paper are as follows:

- To provide a comprehensive study on various malware families along with their impact on the virtualization environment.
- To describe various distinct dynamic features of malware along with the description of output feature log generated by open-source tool's plugins (such as POOLMON, SYSCALL, etc.).
- To provide the case study on the use of an open-source deep memory introspection tool for malware log generation (having file attributes, process attributes, system call attributes, object attributes, etc.) outside the VM.

The rest of the paper is organized as follows: Section 2 gives a short description of malware that has been used for the case study and its impact on virtualization. Section 3 describes the dynamic features of malware that have been extracted using *DRAKVUF*. Section 4 provides the case study in which we have discussed the steps to trace malware. Finally, we have concluded our paper in Section 5.

2 RELATED WORK

Tien et al. [12] proposed a new sandbox architecture that uses memory forensic methods to deliver an agentless, hypervisor-independent sandbox solution. Memory forensics, according to the authors, is an effective method for identifying malware with a high detection rate. Their approach introduced an agentless solution, which solved the issue of VM hypervisor types' high dependency. This paper lacks in providing a detailed analysis of malware.

Huseynov et al. [2] provided a detailed study on providing VM security against keylogger threats. Authors have incorporated the artificial immune system (AIS) technique and VM introspection (VMI) technology to detect anonymous activities. Their approach tracks the events such as system calls, interruption, network activities, etc., and describes how the malicious software can stop any anti-malware software in the system. The paper only considers three dynamic features such as keyboard tracking API and network API, which may not be sufficient for advanced malware analysis.

²<https://drakvuf.com/>

³<https://github.com/volatilityfoundation/volatility>

⁴<https://github.com/libvmi/libvmi>

Table 1: Malware family, type and features

Name	Type	Features
Adload	Trojan Downloader	Downloads files; disable windows proxy setting
Agent	Trojan Downloader	Downloads files to forge security applications.
Renos	Trojan Downloader	Downloads forge security files; asks money to remove malwares.
Vundob	Trojan Downloader	Pops up advertisement; downloads malicious files.
OnLineGames	Password Stealer	Traces user keystrokes; steals user login information.
Lolyda	Password Stealer	Collects the information of the user as well as the system; disables the security services of a system
Zbot	Password stealer	Downloads ransomware in the infected system; disables the firewall and browser security.
Rbot	Backdoor Trojan	Allows attackers to control the infected system; enables attackers via the backdoor to access the data.
Cybot.G	Backdoor Trojan	Gives unauthorized access and control of an infected system to the attacker
Zegost	Backdoor Trojan	Permits the backdoor access and control of the system to the attacker
Hotbar	Adware	Installs a toolbar on the browser; monitors the activity of victim.
Startpage	Trojan	Changes the homepage of the internet browser
Alureon	Trojan	Steals data; gives access of the infected system to attacker.
BHO	Trojan	Displays advertisement; collects user data.
Ceelinject	VirTool	Uses obfuscation techniques; used by other malwares to avoid detection.
Obfuscator	VirTool	Uses techniques to avoid detections and analysis
Delfinjec	VirTool	Steals data and send it to the attacker
FakeRean	Rogue	Acts like antivirus software; offers paid service to clean fake infections
Winwebsec	Rogue	Targets Windows users; displays false security issues; asks fee to fix it.
Vobfus	Worms	Uses obfuscation techniques to avoid detection; spreads through removable drives.

Kapil and Mishra [3] provided a review on various VMI-based security approaches along with various open-source introspection libraries to secure the VMs from multiple attacks. Authors have described various VMI-based applications such as intrusion detection & prevention, memory forensics, malware analysis, etc. Various research challenges are also being provided to give future directions. The paper lacks in providing any practical implementation of the VMI tool/approach.

Our proposed work uses the same VM introspection technique to perform detailed malware analysis by tracking events using *DRAKVUF* plugins. Further, we have provided a detailed overview of the different features of these plugins. Also, we have described how the malicious software uses the system calls.

3 DESCRIPTION OF MALWARE FAMILY

In this section, we have considered various popular malware families specific to windows operating system based on the dataset collected from San José State University, California. The same malware dataset has been considered for analysis purpose[refer section 4] for case study] in the virtualization environment.

Renos

Renos family is a prevalent trojan downloader malware that may download anti-spyware without the user's consent in the system. This software gives false alerts and claims that the system is infected by spyware and asks fee to remove the spyware. If the VM is infected with Renos malware, it will install a system component called a dynamic link library (DLL) in the Windows system folder. The malware may create instability in the VM⁵.

Adload

Adload malware is a type of trojan downloader created in Visual Basic. Botnets are used to produce several adload versions. It downloads an executable file that, in turn, downloads malware. Adload saves executable files on the cloud; therefore, the file's functionality can alter at any time. When an adload is injected into a virtual machine, it may deactivate the window proxy configuration, leaving the VM exposed.⁶

⁵<https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Win32%2FRenos/>

⁶<https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=TrojanDownloader%3AWin32%2FAdload>

Agent

If an agent is injected into the VM, a remote attacker may be able to access the victim's private data. It can also change the settings of windows and edit and erase the victim's data. The variants of agent malware may spread and infect other *guests*⁷.

Browser Help Object

Browser Help Object (BHO) belongs to the trojan type, which may install by other trojan downloader malware in the victim's web browser. BHO is a dynamic link-layer executed whenever a web browser runs. On infecting the VM, It may observe the connections made to different websites, delivers random advertisements and redirect the search queries, and gather the private information of *user*⁸.

Startpage

Startpage is a trojan-type malware that is written in JavaScript. It obfuscates itself to avoid detection and changes the homepage of the web browser without the user's consent. It may perform malicious activities once it is installed in VM. It may change the homepage of the browser without the knowledge of the victim and block security sites. It may connect to a remote host and steals confidential *information*.⁹

CeeInject

CeeInject is a VirTool type malware that uses obfuscation techniques to deceive the anti-virus software. It is used by other malware to avoid detection from the security software. If CeeInject enters in VM, it may obfuscate bitcoin miners that use the computational power of VM to mine bitcoins without users' consent. It may self-mutate and disable the firewall. It connects to unauthorized sites and *malware*.¹⁰

Aleuron

Aleuron family is a data-stealing trojan that steals not only confidential information but also gives unauthorized access to attackers to collect data present on the system. Once installed in VM, it may disable anti-virus software and block windows updates. Aleuron may disable or clear the web browser proxy settings. Some variants of the aleuron family may corrupt the driver *files*.¹¹

FakeRean

FakeRean family is a rogue malware that disguises itself as an anti-virus tool. It is installed by exploit kits or other malware. The malware pretends to scan files and gives false alarms of malware by displaying randomly chosen files. It offers paid service to remove

the allegedly scanned virus. If VM gets infected by fakeRean, it may change the security settings and close the running program *itself*.¹²

Obfuscator

Obfuscator malware is a popular virtual that uses techniques like encryption, anti-debugging, compression and anti-emulation to avoid detection and analysis from anti-virus software. These techniques may use on other malware. It may record the keystrokes and activity of the victim and send information about the infected VM to a remote attacker. It may also give access to an infected VM to an *attacker*.¹³

Rbot

Rbot family is a member of backdoor trojans malware which gives access to control infected systems. After a VM is infected, rbot allows attackers to control the infected system through the IRC server. Attackers may send commands through a special channel, and these commands may instruct rbot to spread to other VMs. It may allow attackers to perform a backdoor function such as denial of service (DoS) *attacks*.¹⁴

Vobfus

Vobfus is a worm-type malware that is written in visual basics for windows users. It spreads through removable drives and by other malware. On infiltrating the VM, it may download malicious files and execute them. Vobfus may affect the performance of an infected VM. It changes the settings of windows and may add VM to botnets. The latest member of Vobfus may exploit a zero-day vulnerability (CVE-2010-2568) in *Windows*.¹⁵

Vundo

Vundo family is a category of trojan downloader malware that displays pop-up advertisements. It can be installed by other malware or can be spread using emails. It may degrade the performance of the VM by disabling the task manager and windows from the safe boot. Vundo may cause a denial of services (DoS) with websites such as Facebook and Google or the blue screen of death. It may infect other VMs through network *drives*.¹⁶

Winwebsec

Winwebsec belongs to rouge malware that is written in JavaScript. It is installed by malware or can be spread through social engineering. It targets Windows users by displaying false alerts of security issues in a system. After infecting the VM, it may ask paid services to remove fictitious malware. It may modify the setting of windows and blocks websites without the user's *consent*.¹⁷

⁷[https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Win32/\\$2FAgent](https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Win32/$2FAgent)

⁸<https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Trojan:Win32/BHO.BO>

⁹<https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Trojan:Win32/Startpage&threatId=15435>

¹⁰<https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=VirTool:Win32/CeeInject.ANO!bit&ThreatID=2147733678>

¹¹<https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Virus:Win32/Alureon.A>

¹²<https://www.microsoft.com/security/blog/2010/05/25/what-is-win32fakearean/>

¹³[https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Win32/\\$2FObfuscator](https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Win32/$2FObfuscator)

¹⁴<https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Backdoor:Win32/Rbot>

¹⁵[https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?name=win32/\\$2Fvobfus](https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?name=win32/$2Fvobfus)

¹⁶[https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Win32/\\$2FVundo](https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Win32/$2FVundo)

¹⁷[https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Win32/\\$2FWinwebsec](https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Win32/$2FWinwebsec)

DelfInject

DelfInject family is one of the widespread virtual malware that is infused into the system through spam emails. The malware uses obfuscation techniques to protect itself from detection and analysis. The malware may be encrypted or compressed and stored inside another file that decodes and loads it. On infusing the VM, the malware steals sensitive, confidential information and sends it to the attacker.¹⁸

Lolyda

Lolyda is a password stealer malware that collects the information of the user as well as the system and sends it to its remote server. Once the lolyda infuses the VM, it may monitor the network activity of the VM to steal the user credential. It may also disable and terminate the security process. It connects with a remote server using a UDP connection and downloads arbitrary files on Windows. It may take a screenshot of the infected VM and forward it to the attacker.¹⁹

OnlineGames

The OnlineGames malware family is a form of password stealer malware that spreads via spam emails or exploits system flaws. The virus may inject malicious DLL files into processes once it has been injected into the VM. It monitors user keystrokes and takes online gaming login information. The attacker receives the information gathered.²⁰

Zbot

Zbot family is one of the password stealer malware that can be installed via spam email or exploit kits like a blackhole. Zbot may download malware in the infected VM that may encrypt the files and asks for money to decrypt them. It shares the financial information and gives the attacker access to and control of the infected VM. Moreover, it can disable the firewall and browser security.²¹

Cycbot.G

Cycbot.G is a backdoor malware that gives unauthorized access and control of an infected VM to the attacker. It connects to an IRC server and receives commands from attackers through a channel. It can be spread to other VMs by exploiting vulnerabilities or backdoor ports opened by other malware. The attacker can perform a DDOS attack using it and retrieves information from the infected VM.²²

Zegost

Zegost is a widespread backdoor malware that may download malicious software without user consent. It may permit backdoor access and control of the VM to the attacker. Zegost may register itself

¹⁸<https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=VirTool:Win32/DelfInject.AN>

¹⁹[https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Win32/\\$2FLolyda](https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Win32/$2FLolyda)

²⁰<https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?name=win32/onlinegames>

²¹[https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=PWS\\$%\\$3AWin32\\$%\\$2FZbot](https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=PWS$%$3AWin32$%$2FZbot)

²²<https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Backdoor:Win32/Cycbot.G>

as a system service for automatic execution. It may connect to a remote host through which attackers can delete, download or run the files, steal sensitive data, and modify windows settings.²³

Hotbar

Hotbar family is a type of adware malware that installs a toolbar on the browser that monitors the activity of a targeted user. Using this toolbar, Hotbar delivers targeted ads on web pages. In addition, it installs other adware without the user's consent. Once Hotbar infects the VM, it installs graphical skins for the web browser. It may collect data, download and executes files from the remote server.²⁴

4 TAXONOMY OF ADVANCED HYBRID FEATURES FOR DYNAMIC ANALYSIS

In this section, we have explained various advanced hybrid features useful for malware analysis. The features have been retrieved using various plugins associated with a deep memory introspection tool, i.e., DRAKVUF [4]. Let us now explain each plugin and its associated features.

4.1 File-Attribute Related Features

The *FILETRACER* plugin is used with introspection tool that uses the "FILEOBJECT" structure with the help of system calls (used by kernel drivers. Various important features can be retrieved through this plugin, such as file names being accessed on the system, file-related system calls being invoked by the malware, and time associated with specific file-related activity (read/write/open/create, etc.). Let us now discuss each feature related to the file-specific information: File manipulation related system calls: The file specific system calls such as *NtWriteFile*, *NtOpenFile*, *NtCreateFile*, *NtQueryAttributesFile* etc. are useful in deriving various important features. For example - frequency, pattern, occurrence, name and/or transition probability of such system calls play a very important role in malware analysis. Filename: The filename is an important attribute/feature during malware log feature extraction. In addition, the file tracer plugin helps to derive other features as well as pattern of files and occurrences of files being accessed, etc.

Time: The time feature helps to know the total time taken by malware while doing specific file-related activities such as read/write/open/create etc. The malware programs usually take longer while accessing or modifying any file.

Log example: A portion of log (specific to file attributes) generated on the execution of a ransomware malware sample in our lab is shown below:

[FILETRACER]

TIME:1641454692.404842 NtCreateFile FileName:"C:\Users\John\Downloads\malware-samples-master\f4d01.bin\taskdl.exe"

Explanation: WNCRYT extension is created as part of the ransomware's encryption process. Once the encryption process is completed, WNCRYT files rename themselves to WNCRY files. This system call is called 50 times in this log.

²³<https://www.microsoft.com/en-us/wdsi/threats/threat-search?query=trojan:win32/zegost!rfn>

²⁴[https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Adware\\$%\\$3AWin32\\$%\\$2FHotbar](https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Adware$%$3AWin32$%$2FHotbar)

4.2 Object Allocation related Features

The *POOLMON* plugin is used to keep track of *ExAllocateWithTag* function which allocates objects on the kernel heap in Windows. Each object is associated with the following three attributes: pool type, number of bytes, and tag values of the allocated object.

Log example: A sample log example of *POOLMON* plugin is given below:

[POOLMON]

ExAllocatePoolWithTag Tag:"CcSc" Type:"NonPagedPool" Size:504

Explanation: On allocation of each object, the following features are assigned to each object: Tag: "CcSc", Type: "NonPagedPool", size 504 bytes. These features are essential for malware analysis. The frequency of occurrence of the *POOLMON* (derived feature) tag signifies how many times objects have been allocated and hence, it is very important feature.

4.3 Process-Attribute Related Features

The *PROCMON* plugin is a downloadable utility for the Microsoft Operating System that captures and displays process activities and has been injected as a plugin with *DRAKVUF*. This plugin can obtain some essential features, such as system calls invoked for process control, process names and time of execution, etc. Process control related system calls: The process control related system calls such as *NtOpenProcess*, *NtCreateUserProcess*, *NtDelayExecution* etc. are useful for analysis purpose and acts as a basic feature. Various other important features can be derived through such log such as frequency of occurrence of system calls, pattern of system calls, transition probabilities of system calls etc.

Process names: The process name is an important basic attribute/feature during malware log feature extraction. A process specific logs can be extracted by setting filtering criteria as 'process name'. The execution tracing of each process is helpful for further analysis.

Time:

The time feature helps to know the total time taken by malware process while doing an intended operation such as delay an execution, create a process, delete a process etc.

Log example: A sample log example of *PROCMON* plugin is given below:

[PROCMON]

TIME:1641454690.589318

NtOpenProcess

ClientName:"\\Device\\HarddiskVolume2\\Users\\John\\Downloads\\malware-samples-master\\Ransomware\\Wannacry\\Wann

acry\\01bf5f10716618404804396e5babe8e080e41aa.bin\\taskhost.exe"

Explanation: This system starts the process of taskhost with a system call invocation *NtOpenProcess*. Windows use taskhost as a host for all processes running from DLLs. It is manipulated by malware to load its entities to the system files to take privilege over the operating system.

4.4 Kernel Execution Tracing Related Features

The *SYSCALL* plugin is in charge of monitoring the execution of function-entry-points that manage Windows system calls. It traps the function whose name start with NT. Let us understand some important features:

Sequence of System calls: During the execution of a program, an execution trace is created, which is one of the crucial features and signifies the pattern or order of various operations (file-related/process control related/object-related, etc.) being invoked by a program. A malware program generate a different pattern of operations than a benign program. For example, multiple failed attempts to a remote connection will result into series of Login Attempt Failed related system calls.

Frequency/transition probabilities of system calls: The execution tracing is helpful in deriving some important features using feature extraction approaches. For example, frequency of each unique system call (Bag of system calls [5]), frequency of each short system call trace (Bag of n-gram approach [5]), transition probabilities of system calls (Conflow Flow Graph approach [5]) are important derived features.

Log example: Let us consider a sample log of an execution trace of a process, given below:

[SYSCALL]

ransomware.exe PID: 2343, PPID: 4532

NtQueryAttributesFile, NtDelayExecution, NtOpenFile, NtQueryVolumeInformationFile, NtClose, NtWriteFile, NtQueryInformationProcess, NtCreateFile, NtOpenKeyEx, NtRequestWaitReplyPort, NtMapViewOfSection

Explanation: Here, it can be seen that *SYSCALL* plugin has provided a sequence of file and process related system calls in certain order, invoked by process *ransomware.exe* having PID: 2343. The execution tracing log depicts the operations performed by *ransomware.exe* to access the user, system files and modify the registry values of system to take control over whole operating system.

4.5 Object Creation Related Features

The plugin can be used to keep track of the record of the number of times common objects have been created in Windows. This is done by tracking the invocation of *ObCreateObject*.

Log example: A sample log example of *OBJMON* plugin is given below:

TIME:1641454674.924380

ObCreateObject

filename:"\\Device\\HarddiskVolume2\\Users\\John\\Downloads\\malware-samples-master\\071661840480439c6e5babe8e080e41aa.bin\\wannacry.exe"

Explanation: In this log, the useful data is the occurrence of the *ObCreateObject* system call indicating the creation of a common object in Windows.

5 CASE STUDY

A malware introspection environment has been set up on Xen Server using *DRAKVUF*²⁵ which is a tool for introspection and an agentless black-box binary analysis system designed for virtualization. Without needing to install any extra software within the VM which is used for analysis, *DRAKVUF* provides for in-depth execution tracing of arbitrary binaries (including operating systems) [4]. The system specifications are: UBUNTU LTS 20.04 OS (physical machine) having Intel i5 processor, 16 GB RAM, and 256 GB SSD. Specifications of VM are - Windows 7 Ultimate OS, 3 GB

²⁵<https://drakvuf.com/>

RAM, 50 GB SSD, 2 CPU cores. A case study is demonstrated below to better understand the malware log generation process in the analysis environment, set up in our lab:

Step 1 - Installation of Xen hypervisor

Xen hypervisor [10] is installed on the top of the testbed setup considered, and Dom0 VM is configured with 8 GB RAM and 4 vCPU cores. The command to verify the Xen installation is given in Figure 1.

```
cs13@cs13-HP-280-G2-MT-Legacy:~$ sudo xen-detect
Running in PV context on Xen V4.16.
cs13@cs13-HP-280-G2-MT-Legacy:~$ sudo xl list
```

Name	ID	Mem	VCPUs	State	Time(s)
Domain-0	0	8096	4	r-----	3631.0

Figure 1: Domain-0 is Xen Server

Step 2 - Creation of Logical Volume Manager (LVM) group

LVM is a device mapper framework that helps in creating logical partitioning of the volume/storage to be assigned to a VM. pvcreate /dev/sdb2 : Created a physical volume on the free space of the disk.

vgcreate vg /dev/sda2 : A volume group is created, which gathers logical and physical volume into one administrative unit.

lvcreate -L80G -n windows7-sp1 vg : A logical volume is created in the LVM.

Step 3 - Configuration of the network bridge

Control utilities for Linux bridging are installed, and the Debian network interface is configured. Bridge name and network interface name are provided to the configuration file as shown in Figure 2 that helps in adding VMs to our external network.

```
auto lo
iface lo inet loopback

auto enp4s0
iface enp4s0 inet manual

auto virbr0
iface virbr0 inet dhcp
bridge_ports enp4s0
```

Figure 2: Setting Linux bridge

Step 4 - Creation of Windows Configuration file

A configuration file is created for Windows, in our case Windows7, and the file is saved in the linux directory, which contains Xen Configuration files, i.e. "/etc/xen". This configuration helps in creating the VM.

Step 5 - Installing windows

Windows 7 VM is created and started as shown in Figure 3. The VM is booted using the command "gynviewer localhost". Windows is installed. The specs of the VM will be the same as defined in the configuration file.

```
cs13@cs13-HP-280-G2-MT-Legacy:~$ sudo xl create /etc/xen/win7.cfg
Parsing config from /etc/xen/win7.cfg
cs13@cs13-HP-280-G2-MT-Legacy:~$ sudo xl list
```

Name	ID	Mem	VCPUs	State	Time(s)
Domain-0	0	8096	4	r-----	4607.9
windows7-sp1	6	3000	1	-b-----	0.0

Figure 3: Creation of VM

Step 6 - Building LibVMI

LibVMI folder, which comes in handy with the Drakvuf folder, is built. All C programs with python bindings are installed. LibVMI helps in easy monitoring of running VM by viewing its memory, hardware events, and vCPU [13].

Step 7 - Creation of JSON and LibVMI config

A JSON profile of the VM (windows7) is created by providing the PDB Guid and kernel filename. Further, using this profile, the LibVMI config file is created. LibVMI is ready to monitor the guest VM as shown in Figure 4

```
Process listing for VM windows7-sp1 (id=1)
[ 4] System (struct addr:fffffa80023c7860)
[ 252] smss.exe (struct addr:fffffa8002c9ab00)
[ 328] csrss.exe (struct addr:fffffa8003415710)
[ 376] csrss.exe (struct addr:fffffa80023cd060)
```

Figure 4: Checking LibVMI by running process list

Step 8 - Installing Drakvuf

Drakvuf is built and installed in the specified directory, i.e., "drakvuf" and all the executable files and programs are built from source code. Drakvuf will now trace the system execution as shown in Figure 5 & 6.

```
cs-13@cs13-HP-280-G2-MT-Legacy:~/drakvuf$ sudo ./src/drakvuf -r /root/windows7-sp1
.json -d 3 -x socketmon -t 60 -i "B:\virus1.exe" | grep -w "virus1.exe" > virus1.txt
1647673598.507425 DRAKVUF v1.0-git20220122081915+d481082-1 Copyright (C) 2014-2022
Tamas K Lengyel
```

Figure 5: DRAKVUF command to generate logs

ACKNOWLEDGMENTS

The Science and Engineering Research Board, Department of Science and Technology (SERB-DST) provided funding for this study under Project File No. SPG/2021/002003. The authors would like to express their gratitude to SERB-DST for their intellectual generosity and research assistance.

```
[SYSCALL] TIME:1647674213.220070 VCPU:0 CR3:0x4f154000 "\\Device\\HarddiskVolume2\\Windows\\explorer.exe":NtCreateFile SessionID:1 PID:1564 PPID:1428 Module:"nt" vCPU:0 CR3:0x4f154000 Syscall:82 NArgs:11 FileHandle:0x6A6DC38 DesiredAccess:0x00000000 ObjectAttributes:"??\\B:\\virus1.exe" IoStatusBlock:0x6A6DBD0 AllocationSize:0x0 FileAttributes:0x80 ShareAccess:0x1 CreateDisposition:0x1 CreateOptions:0x10000 EaBuffer:0x0 EaLength:0x0
[FILETRACER] TIME:1647674213.220441 VCPU:0 CR3:0x4f154000 "\\Device\\HarddiskVolume2\\Windows\\explorer.exe":NtCreateFile SessionID:1 PID:1564 f
```

Figure 6: Generated Logs

6 CONCLUSION

With the emerging popularity of virtualization, the risk of compromising the security and privacy of end users has also increased. Malicious users exploit the vulnerabilities of virtual components and cause millions of losses to organizations. In this paper, we have provided a study on advanced malware and its impact on the virtual environment. A case study has been performed using the open-source deep memory introspection tool (DRAKVUF & LibVMI) to extract the dynamic features from the VM at the hypervisor. We have thoroughly explained these dynamic features of malware extracted using the DRAKVUF plugins such as PROCMON, OBJMON, SYSCALL, POOLMON, etc. In the future, we will extend the malware feature extraction with an efficient detection algorithm.

REFERENCES

- [1] Luca Cavignone, Michal Choras, Igino Corona, Artur Janicki, Wojciech Mazurczyk, Marek Pawlicki, and Katarzyna Wasielewska. 2021. Tight Arms Race: Overview of Current Malware Threats and Trends in Their Detection. *IEEE Access* 9 (2021), 5371–5396. <https://doi.org/10.1109/ACCESS.2020.3048319>
- [2] Huseyn Huseynov, Kenichi Kourai, Tarek Saadawi, and Obinna Igbe. 2020. Virtual Machine Introspection for Anomaly-Based Keylogger Detection. In *2020 IEEE 21st International Conference on High Performance Switching and Routing (HPSR)*. 1–6. <https://doi.org/10.1109/HPSR48589.2020.9098980>
- [3] Divya Kapil and Preeti Mishra. 2021. *Virtual Machine Introspection in Virtualization: A Security Perspective*. Association for Computing Machinery, New York, NY, USA, 117–124. <https://doi.org/10.1145/3474124.3474140>
- [4] Tamas K. Lengyel, Steve Maresca, Bryan D. Payne, George D. Webster, Sebastian Vogl, and Aggelos Kiayias. 2014. Scalability, Fidelity and Stealth in the DRAKVUF Dynamic Malware Analysis System. In *Proceedings of the 30th Annual Computer Security Applications Conference (New Orleans, Louisiana, USA) (ACSAC '14)*. Association for Computing Machinery, New York, NY, USA, 386–395. <https://doi.org/10.1145/2664243.2664252>
- [5] Preeti Mishra, Palak Aggarwal, Ankit Vidyarthi, Pawan Singh, Baseem Khan, Hassan Haes Alhelou, and Pierluigi Siano. 2021. VMShield: Memory Introspection-Based Malware Detection to Secure Cloud-Based Services Against Stealthy Attacks. *IEEE Transactions on Industrial Informatics* 17, 10 (2021), 6754–6764. <https://doi.org/10.1109/TII.2020.3048791>
- [6] Preeti Mishra, Akansh Gupta, Palak Aggarwal, and Emmanuel S. Pilli. 2022. vServiceInspector: Introspection-assisted evolutionary bag-of-ngram approach to detect malware in cloud servers. *Ad Hoc Networks* (2022), 102836. <https://doi.org/10.1016/j.adhoc.2022.102836>
- [7] Preeti Mishra, Emmanuel S. Pilli, Vijay Varadharajan, and Udaya Tupakula. 2017. Intrusion detection techniques in cloud environment: A survey. *Journal of Network and Computer Applications* 77 (2017), 18–47. <https://doi.org/10.1016/j.jnca.2016.10.015>
- [8] Preeti Mishra, Emmanuel S Pilli, Vijay Varadharajan, and Udaya Tupakula. 2017. VAED: VMI-assisted evasion detection approach for infrastructure as a service cloud. *Concurrency and Computation: Practice and Experience* 29, 12 (2017), e4133.
- [9] Preeti Mishra, Emmanuel S. Pilli, Vijay Varadharajan, and Udaya Tupakula. 2016. NvCloudIDS: A security architecture to detect intrusions at network and virtualization layer in cloud environment. In *2016 International Conference on Advances in Computing, Communications and Informatics (ICACCI)*. IEEE, 56–62. <https://doi.org/10.1109/ICACCI.2016.7732025>
- [10] Preeti Mishra, Vijay Varadharajan, Emmanuel S. Pilli, and Uday Tupakula. 2020. VMGuard: A VMI-Based Security Architecture for Intrusion Detection in Cloud Environment. *IEEE Transactions on Cloud Computing* 8, 3 (2020), 957–971. <https://doi.org/10.1109/TCC.2018.2829202>
- [11] Camille Singleton, Charles DeBeck, Joshua Chung, McMillen, Craig Dave, Moore Scott, and Scott. 2022. *The Tenth Annual X-Force Threat Intelligence Index*. Technical Report.
- [12] Chin-Wei Tien, Jian-Wei Liao, Shun-Chieh Chang, and Sy-Yen Kuo. 2017. Memory forensics using virtual machine introspection for Malware analysis. In *2017 IEEE Conference on Dependable and Secure Computing*. IEEE, 518–519.
- [13] Haiquan Xiong, Zhiyong Liu, Weizhi Xu, and Shuai Jiao. 2012. Libvmi: a library for bridging the semantic gap between guest OS and VMM. In *2012 IEEE 12th International Conference on Computer and Information Technology*. IEEE, 549–556.
- [14] Lifan Xu, Dongping Zhang, Nuwan Jayasena, and John Cavazos. 2018. HADM: Hybrid Analysis for Detection of Malware. *Lecture Notes in Networks and Systems*, 702–724. https://doi.org/10.1007/978-3-319-56991-8_51
- [15] Hancui Zhang and Weida Zhou. 2022. A two-stage virtual machine abnormal behavior-based anomaly detection mechanism. *Cluster Computing* 25, 1 (2022), 203–214.